

SQL EXCEL PROJECT REPORT

Company Analysis Database

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Tools Used: MySQL Workbench, SQL, Excel/CSV

1. Project Overview

- Developed a relational **Company Analysis Database** using four CSV source files.
- Converted raw CSV data into structured MySQL tables for customer, product, order and sales analysis.
- Performed data cleaning, validation, database design, constraint implementation and business analysis.

2. Source Data

Source File	Purpose	Records
customers.csv	Customer information	220,000
products.csv	Product information	120,000
orders.csv	Order information	420,000
sales.csv	Sales transactions	500,000

3. Database Design

Database: companyanalysis

Tables:

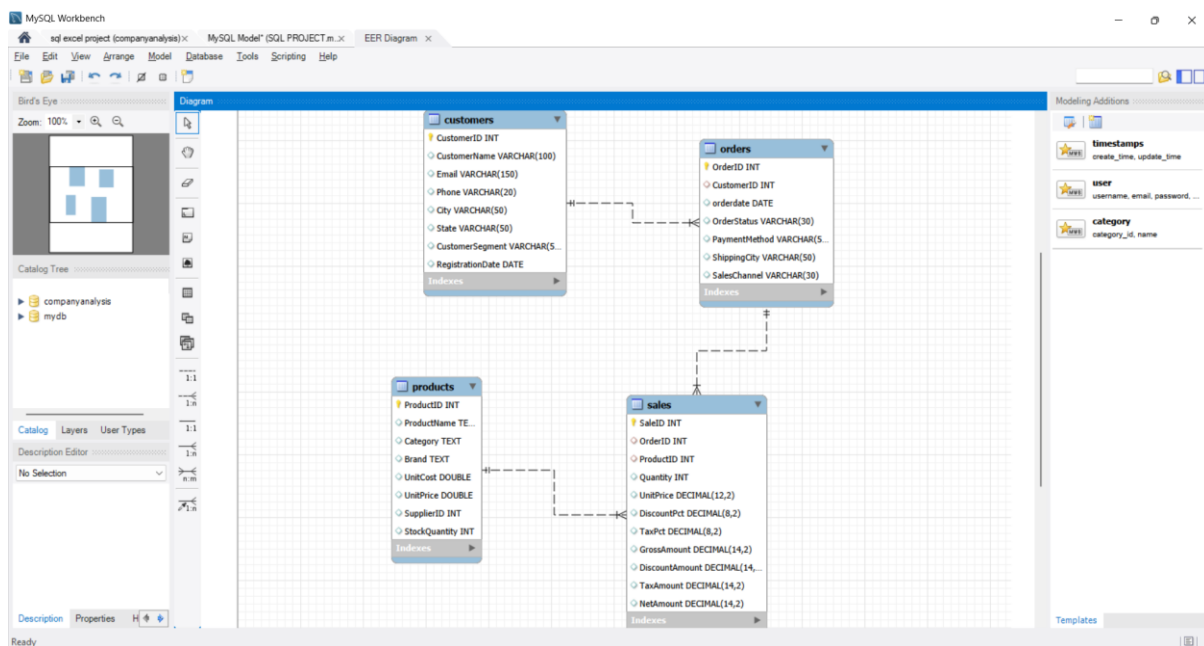
- customers — customer details
- products — product details
- orders — order information
- sales — transaction/line-item information

Relationships:

- customers.CustomerID → orders.CustomerID — 1:N
- orders.OrderID → sales.OrderID — 1:N
- products.ProductID → sales.ProductID — 1:N

Design: 1NF, 2NF and 3NF principles followed.

EER Diagram



4. ETL & Data Quality

- Imported all four CSV files into MySQL using load data local infile.
- Created staging tables and production tables.
- Converted date fields to MySQL DATE format.
- Validated duplicate records, duplicate IDs, NULL values, invalid IDs and orphan records.
- Validated numeric ranges, negative values, zero values, category consistency and outliers.
- No valid source records were removed or altered.
- Future-dated orders were identified and retained as intentional future-dated records.

5. Database Constraints & Integrity

Implemented and tested:

- Primary Key constraints
- Foreign Key constraints
- UNIQUE constraint on customer email
- NOT NULL constraints
- CHECK constraints for quantity, price, discount and tax
- DEFAULT order status
- Referential integrity and orphan-record checks

6. SQL & Business Analysis

SQL concepts demonstrated:

- SELECT, WHERE, ORDER BY, DISTINCT
- CASE statements
- GROUP BY and HAVING
- INNER JOIN, LEFT JOIN and multi-table JOINS
- Subqueries and CTEs
- UNION
- Window functions
- Analytical queries

Business analysis included:

- Total revenue
- Order, customer and product-level revenue
- Monthly revenue
- Top customers and products
- Customer segment analysis
- Payment method and sales channel analysis
- Product/category performance

7. Validation & Performance

- Revenue, discount, tax and net amount calculations were independently validated.
- Rounding differences in discount/tax calculations were investigated and found to be within ₹0.01.
- Final row counts reconciled with source data.
- Data-quality checks completed successfully.
- Indexes implemented and query execution plans reviewed using EXPLAIN.
- Before/after indexing performance was reviewed.

8. Final Outcome

The project successfully transformed four raw CSV datasets into a structured and validated MySQL relational database. The final database maintains data integrity through appropriate keys and constraints and supports SQL-based business and financial analysis.